

Device-level parasitic emission as a routing constraint in quantum networks: a VOA electroluminescence case study

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<https://github.com/Sguy1908/QiOpt4QNet-Simulator>

Abstract

Entanglement-routing optimizers assume that a link is fully described by a fidelity, a generation probability and a capacity. Hardware side channels break that abstraction: Li *et al.* recently showed that the forward-biased $p-i-n$ variable optical attenuators (VOAs) in chip-based QKD transmitters emit spontaneous electroluminescence near 1107 nm, which acts as a passive Trojan-horse channel and as an independent noise source at the receiver. Here we implement their analytic model, check it against the numbers they publish, and couple it to a quantum-network simulator so that the same allocators used for routing can be evaluated under the device effect. The model reproduces the published mean photon numbers (0.0048, 0.0388, 0.0977), an ideal reach of ≈ 320 km and a $\approx 50\%$ short-range key-rate loss in the worst-case operating point. At network level the effect is dominated not by fidelity loss (mean $\Delta F \leq 0.011$ per link, $\leq 6.1\%$ utility) but by security: with a 2.0 V, 1.6 ns drive, a security-blind exact allocator admits requests of which 6.0 of 7.45 traverse links whose passive-THA key rate is zero, and a security-aware allocator keeps only 2.0 of 7.75 requests. A 10 dB band-pass filter on the emission removes the penalty. We also find that greedy allocators reorder their picks under 10^{-6} -scale fidelity perturbations. The device model is a reconstruction of the published analysis, not a new measurement.

Introduction

Quantum networks distribute entanglement across repeater chains and grids, and the routing layer decides which path and purification schedule each request receives under memory and link-capacity limits. The optimizers studied in this repository—an exact CP-SAT reference, greedy heuristics, a Metropolis annealer, tensor-network compression and learned rankers—all consume a per-link summary: raw fidelity, success probability, latency and capacity.

Implementation security is largely absent from that summary. Chip-based QKD is usually analysed at the protocol level with idealised devices, yet integrated photonic components combine classical and quantum functions in a tight footprint. Li *et al.* [1] identified one such mechanism: a carrier-injection VOA is a lateral $p-i-n$ diode, so the recombination that produces attenuation also produces light. They measured the emission (centre wavelength inferred at ≈ 1078 – 1093 nm from an on-chip interferometer, close to the 1107 nm silicon band gap), and analysed two consequences: a *passive Trojan-horse attack* (THA) when the VOA sits upstream of the encoder, and a *dual-source flaw* when it sits downstream.

We ask a network-level question the original point-to-point study does not: when links are driven by such transmitters, how much does the device effect change what a routing optimizer should admit, and does it matter whether the optimizer knows about it? Our contributions are (i) a tested implementation of the published gain, QBER and THA-leakage expressions; (ii) a coupling of that model to link fidelity and to a link-level security predicate; and (iii) a controlled comparison of allocators on clean, security-blind and security-aware instances, including the paper’s proposed spectral-filter mitigation.

Results

The device model reproduces the published numbers

The model takes the VOA drive voltage U and the signal pulse width Δt , maps the measured emission count rate $C(U)$ to a mean photon number per pulse, $\mu = -\ln[1 - C(U)\Delta t]$ (Eqs. 1), and propagates it through the gain/QBER and leakage expressions in Methods. Using the two published count rates (Table 1) it returns $\mu_{\text{Eve}} = 0.0048, 0.0388, 0.0977$ for the paper’s three operating points, matching the reported values. Without parasitic light the decoy-state BB84 rate is 0.099 bit/pulse at zero distance and vanishes at 322 km, consistent with the ≈ 320 km reach in Fig. 6 of ref. [1]. At $U = 2.0$ V, $\Delta t = 1.6$ ns the dual-source rate at zero distance falls to 0.048 bit/pulse (a 51 % reduction) and is non-monotonic in distance, rising to 0.056 at 5 km before falling, as in their Fig. 8. The noise-aware rate never exceeds the ideal rate at any distance; the penalty relative to it falls below 5 % by 20 km and to about 1 % at 30 km.

Two caveats. Only two count rates are published, so $C(U)$ is interpolated log-linearly between 1.4 and 2.0 V (and extrapolated above), ramped linearly from zero at the 0.8 V turn-on to the 1.4 V anchor, and set to zero below the turn-on; intermediate voltages are model output, not data. And for the conventional GLLP analysis the maximum passive-THA distances we obtain (75.5, 30.5 and 12.0 km) are of the same order as, but not identical to, those read from their Fig. 6; we did not have their numerical values or code, so we make no claim of exact agreement there.

Table 1. Simulation parameters (from Table 1 of ref. [1] unless marked).

Parameter	Symbol	Value	Note
Misalignment error	e_d	0.0061	
Background yield	Y_0	2×10^{-8}	
Detection efficiency (signal)	η_{Bob}	0.78	
Error-correction inefficiency	f_{EC}	1.2	
Fibre loss, 1550 nm / 1077 nm	α, α'	0.2 / 0.8 dB km $^{-1}$	
Detection efficiency (EL)	η'_{Bob}	0.25	
Signal / decoy intensities	s, ν, ω	0.48, 0.02, 0.001	
$C(1.4 \text{ V}), C(2.0 \text{ V})$		$2.38 \times 10^7, 5.82 \times 10^7$ cps	
Sifting factor	q	1/2	$p_Z \rightarrow 1$; ours
Link length range	L	2–30 km	network study; ours

A noise-blind decoy estimator over-certifies only marginally

Ref. [1] argues that unmodelled parasitic clicks bias decoy-state parameter estimation. We implemented that scenario by feeding the observed (noise-inflated) gains to the standard two-decoy estimator (“naive”) and compared it with a noise-aware bound in which the parasitic source is characterised, its clicks are removed from the decoy statistics, and every noise-induced error is charged to the single-photon key bits (Methods). By construction the aware rate never exceeds the ideal rate. With Table 1 the naive rate is *higher* than the aware one at every distance, in the direction the paper predicts, but by at most 0.74 % (where the rate exceeds 10^{-4} bit/pulse; 0.68 % at 0 km, 0.64 % at 5 km and < 0.01 % at 50 km). The key-rate loss in Fig. 1 therefore comes almost entirely from the added QBER, not from mis-estimation. The paper reports no explicit expression for this case, so this is our reconstruction; we do not carry the estimator bias into the network study.

At network level the dominant effect is security, not fidelity

We coupled the device model to the routing pipeline (Methods). Each instance is a 3×3 grid or an 8-node chain with link lengths drawn from 2–30 km, raw link fidelity 0.92, and 10 requests; we ran 10 seeds per topology and each of the three VOA operating points, with the exact CP-SAT allocator as reference

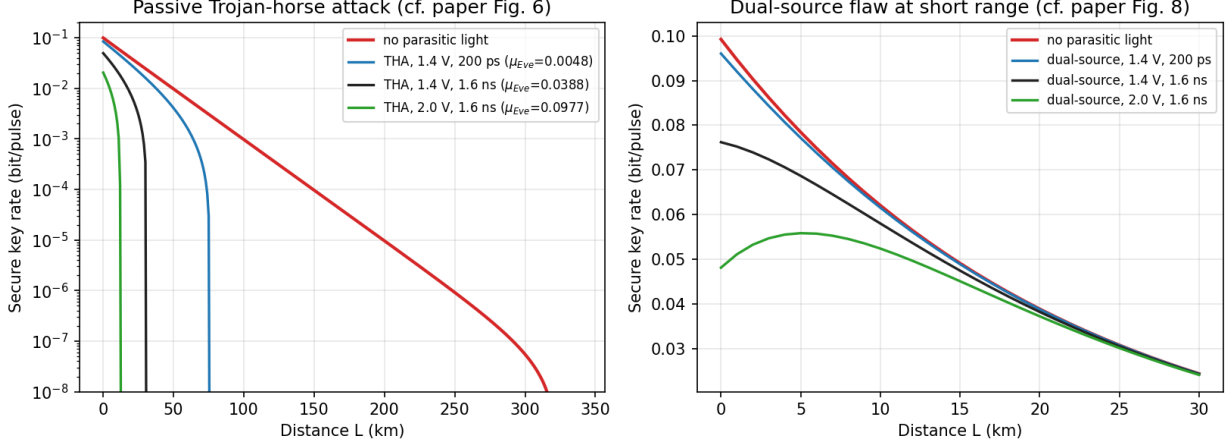


Figure 1. Secure key rate versus distance. Left: passive THA for the three VOA operating points, against the ideal decoy-state BB84 rate (red). Right: dual-source flaw over the first 30 km, using the noise-aware single-photon bound. μ_{Eve} values are in the legend. The curves are our reconstruction of the analysis in ref. [1].

and three greedy heuristics as robustness checks. Each instance is solved three ways: *clean* (no emission), *blind* (degraded links, the optimizer is unaware of the leak) and *aware* (bundles crossing a link with zero passive-THA key rate are removed before allocation).

The fidelity channel is weak. The mean per-link loss is $\Delta F = 0.0005, 0.0042, 0.0105$ for the three operating points, and the exact solver’s utility falls by 0.1 %, 1.7 % and 6.1 % relative to clean (Table 2). It is larger only on short links: an individual 2 km link at 2.0 V, 1.6 ns loses ≈ 0.04 in fidelity, while a 25 km link loses < 0.002 .

The security channel is not weak. In the worst-case configuration 6.15 of the grid/chain links on average exceed the passive-THA reach (12.0 km), and the blind exact allocator’s solution routes 6.0 of its 7.45 admitted requests over at least one such link—requests that it would report as successes while their key is exposed. A security-aware allocator avoids those links and, because they make up a large share of these sparse topologies, can serve only 2.0 of 7.75 requests, a 74 % loss of admitted requests and 73.5 % of utility. Neither of the two milder operating points produces an insecure link at these lengths.

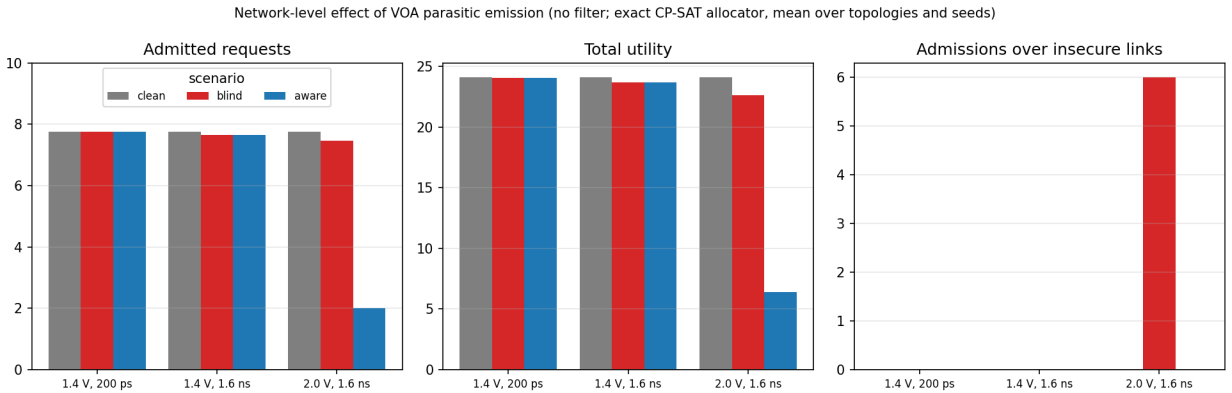


Figure 2. Network-level effect of VOA parasitic emission. Admitted requests (left), total utility (middle) and admissions traversing an insecure link (right) for clean, security-blind and security-aware allocation at the three operating points, without a filter. Exact CP-SAT allocator, mean over both topologies and 10 seeds; the “clean” scenario emits no light, so it has no insecure links.

Table 2. Exact-allocator results without a filter (mean over 2 topologies $\times$ 10 seeds; $n = 20$ per cell).

Operating point	ΔF	clean util.	blind util.	aware util.	blind insecure adm.
1.4 V, 200 ps	0.0005	24.11	24.09	24.09	0.0
1.4 V, 1.6 ns	0.0042	24.11	23.71	23.71	0.0
2.0 V, 1.6 ns	0.0105	24.11	22.63	6.40	6.0

A band-pass filter removes the penalty

Following the mitigation proposed in ref. [1], we model a filter after the VOA as a constant attenuation of the emission, scaling μ_{EL} and μ_{Eve} by $10^{-A/10}$. At the worst-case drive, $A = 10$ dB brings the exact solver back to 24.06 of 24.11 utility with no insecure link, and $A = 20$ dB to 24.10 (Fig. 3). We modelled the filter as a scalar and did not model its insertion loss, bandwidth or the residual out-of-band transmission of a real component.

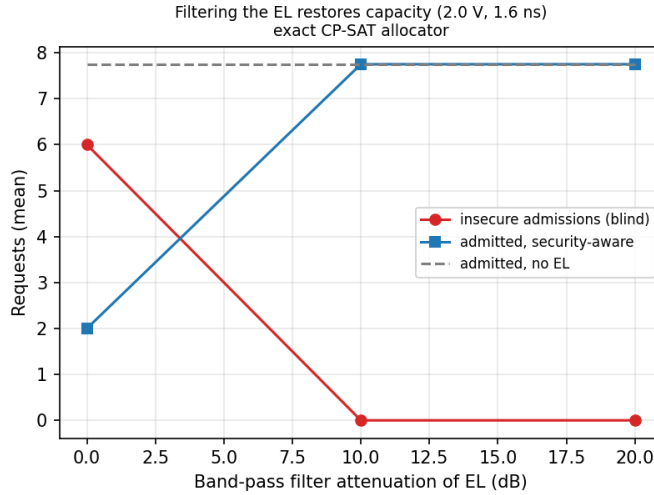


Figure 3. Effect of filtering the emission at 2.0 V, 1.6 ns. Insecure admissions of the blind allocator (red) and requests admitted by the security-aware allocator (blue) versus filter attenuation, against the no-EL level (grey). Only three attenuation values were simulated.

Greedy allocators are brittle to negligible fidelity changes

With a 20 dB filter the mean per-link fidelity change is $\Delta F \approx 5 \times 10^{-5}$ (between 3×10^{-6} and 2×10^{-4} across instances), yet the greedy heuristics still admit 0.1–0.2 fewer requests on average than in the clean scenario, whereas the exact allocator loses none (7.75 vs 7.75). The bundle lists are identical or nearly identical in size; a perturbation of the utility of order 10^{-6} reorders ties and the greedy pick cascade diverges. This is a property of the heuristics, not of the physics, and is why we report the exact allocator as the reference and the per-allocator breakdown separately (`parasitic_network_by_allocator.csv`).

Summary of the routing framework

For context, the surrounding framework (detailed in the companion paper [3]) formulates bundle selection as a QUBO and benchmarks a Metropolis annealer, tensor-network compression and OpenJij against an exact ILP. Reported there: mean 1.5% optimality gap for the Metropolis annealer; a request-conditioned penalty calibration that tightens capacity coefficients on 12.9% of 1,080 instances; chance-constrained fidelity requirements that expose a 5–14% implicit violation rate in the deterministic rule; local-repair recourse

with up to $1065\times$ speedup over replanning; and conflict-aware purification scheduling matching per-request throughput on 85 % of instances at $2.06\times$ lower Bell-pair cost. The present study adds a device-level layer beneath those results.

Discussion

We turned a point-to-point hardware finding into a routing question and found that the useful answer is a security predicate on links, not a fidelity correction. For the two operating points nearest to typical use the effect on the optimizer is small: the fidelity term is at most a few percent and no link is insecure at metropolitan lengths. The worst-case drive matters because the passive-THA reach (12 km) falls inside a plausible metropolitan link distribution; the model then implies that a routing layer that ignores the leak reports secure service that is not secure.

Several limitations bound these statements. (i) All device inputs are taken from a single published study; we did not measure any device, and our reconstruction of the two-decoy estimator and the $C(U)$ interpolation are our own. (ii) The map from QBER to Werner fidelity ($F = 1 - \frac{3}{2} \text{QBER}$) and the choice to treat every link as a BB84-style detector link are modelling assumptions of this repository, not of the paper; entanglement-swapping links with memories would differ. (iii) The “insecure link” predicate uses the worst-case ideal Eve of the paper (unit efficiency, no dark counts), so it is conservative. (iv) The network instances are small (9 and 8 nodes, 10 requests), the length distribution is uniform, and $n = 20$ per cell, so the absolute utility numbers should not be extrapolated; the qualitative split between a small fidelity effect and a large security effect is what we consider robust. (v) The filter is a scalar attenuation.

The main practical recommendation is architectural: carry a per-link security attribute—here a boolean derived from the key rate—through bundle generation, and treat it as a hard constraint in the optimizer, rather than as a fidelity correction after the fact. Extending the same interface to other parasitic emitters (phase modulators, on-chip detectors) is a natural next step, as suggested in ref. [1].

Methods

Emission and leakage model

The mean photon number per signal window is

$$\mu = -\ln[1 - C(U) \Delta t], \quad (1)$$

with $C(U)$ the VOA emission count rate (Table 1); for the passive THA $\mu_{\text{Eve}} = \mu$ under an ideal, dark-count-free unit-efficiency detector. For a coherent copy of the four BB84 polarisations the basis dependence is

$$\Delta = \frac{1}{2} \left[1 - e^{-\mu_{\text{Eve}}} \left(\cosh \frac{\mu_{\text{Eve}}}{\sqrt{2}} + \frac{1}{2} \sinh \frac{\mu_{\text{Eve}}}{\sqrt{2}} \right) \right], \quad (2)$$

and the phase-error bound of the GLLP–Koashi analysis is

$$e'_X = e_X + 4\Delta'(1 - \Delta')(1 - 2e_X) + 4(1 - 2\Delta')\sqrt{\Delta'(1 - \Delta')e_X(1 - e_X)}, \quad \Delta' = \Delta/Y_1, \quad (3)$$

set to $1/2$ if $\Delta' \geq \frac{1}{2}$. The passive-THA key rate is

$$R = q \left\{ Q_1 [1 - h_2(e'_X)] - Q_s f_{\text{EC}} h_2(E_s) \right\}, \quad (4)$$

with $Q_1 = s e^{-s} Y_1^L$, Y_1^L and $e_X = e_1^U$ from the two-decoy estimator [2] applied to the clean gains, and $q = 1/2$.

Dual-source gain and QBER

For signal intensity γ , transmittances η (signal) and η' (parasitic), $s_c = 1 - e^{-\gamma\eta}$ and $n_c = 1 - e^{-\mu_{\text{EL}}\eta'}$:

$$Q = 1 - (1 - Y_0)(1 - s_c)(1 - n_c), \quad (5)$$

$$EQ = Y_0 e_0 + e_d s_c + e_0 n_c - Y_0 e_0 e_d s_c - Y_0 e_0^2 n_c - e_d e_0 s_c n_c + Y_0 e_0^2 e_d s_c n_c, \quad (6)$$

with $e_0 = 1/2$ (the paper's Eqs. 23–24). The exact single-photon yield Y_1 and error mass $E_1 Y_1$ in the presence of noise follow by setting the signal photon number to one in the same construction (Eq. 66 of the paper, averaged over the noise Poisson law).

Noise-aware key rate

The parasitic source is characterised, so its clicks are removed from the decoy statistics and the single-photon contribution $Q_1 = s e^{-s} Y_1^L$ is the same bound as in the noise-free analysis; noise adds errors but no key. Charging all of them to the single-photon key bits,

$$e_1 = \min\left\{\frac{1}{2}, e_1^U + \frac{(E_1 Y_1)_{\mu_{\text{EL}}} - (E_1 Y_1)_0}{Y_1^L}\right\}, \quad R = q\left\{Q_1[1 - h_2(e_1)] - Q_s f_{\text{EC}} h_2(E_s)\right\}, \quad (7)$$

with Q_s, E_s the observed noisy signal gain and QBER. This rate equals the ideal one at $\mu_{\text{EL}} = 0$ and is never larger than it at any distance (a regression test checks 0–340 km at three drive levels and $\mu_{\text{EL}} = 0.3, 1.0$, and the 50 km case explicitly).

Network coupling

For a link of length L , raw fidelity F_0 and $e_d = \frac{2}{3}(1 - F_0)$, let $E(\mu_{\text{EL}})$ be the QBER above at intensity $\gamma = s$. The degraded fidelity is

$$F = \max\left\{0.25, F_0 - \frac{3}{2}[E(\mu_{\text{EL}}) - E(0)]\right\}, \quad (8)$$

so a link without emission is unchanged exactly. A link is *insecure* if the passive-THA key rate at L is 0. A bundle is removed in the security-aware scenario if any of its edges is insecure. A filter of A dB multiplies μ by $10^{-A/10}$.

Instances and statistics

Topologies: 3×3 grid and 8-node chain, edge capacity 4, memory 8, raw fidelity 0.92. Per-link lengths are drawn uniformly from 2–30 km (seeded, identical across scenarios). Requests: 10 per instance, minimum fidelity uniform in $[0.5, 0.9]$, weights uniform in $[0.5, 5]$, three candidate paths per request, purification levels 0–2. Allocators: `cp_sat_exact` (reference), `utility_per_resource_greedy`, `congestion_aware_greedy`, `greedy_local_search`. Seeds 0–9. Values are means over 20 instances per cell. Paired differences (clean minus blind exact-solver utility) are non-zero in all 20 instances at every operating point, with mean $\pm$ standard error 0.02 ± 0.002 , 0.40 ± 0.10 and 1.48 ± 0.35 utility units for the three operating points; we did not run formal significance tests.

Data availability

All CSVs and figures are under `QNet_Sim/results/experiments/`: `parasitic_key_rate.csv`, `parasitic_summary.csv`, `parasitic_network.csv`, `parasitic_network_summary.csv` and `parasitic_network_by_allocator.csv`.

Code availability

Model: `QNet_Sim/src/hardware/parasitic_emission.py` and `QNet_Sim/src/hardware/parasitic_network.py`. Experiment: `QNet_Sim/experiments/run_parasitic_emission.py`. Tests (24): `QNet_Sim/tests/test_parasitic_emission.py` and `QNet_Sim/tests/test_parasitic_network.py`.

Author contributions, competing interests, funding

To be completed by the authors.

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